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Enhancing the wear resistance of AISI 201 alloy by the addition of TiN particles through ultrasonic shot peening

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ABSTRACT

AISI 201 alloy steel is widely used due to its low cost and corrosion resistance, but its low hardness and wear resistance limit its use in high-wear environments. Surface enhancement technologies can improve wear resistance while maintaining cost advantages. In this study, ultrasonic shot peening combined with TiN particles (USG) was utilized to fabricate a TiN coating on the surface of AISI 201 alloy steel. The experimental results revealed that USG treatment induced significant grain refinement, forming an equiaxed nanocrystalline structure and increasing dislocation density. This microstructural evolution contributed to a 157 % increase in cross-sectional hardness. Dry sliding ball-on-disk tests at room temperature showed a 70.6 % reduction in the average coefficient of friction and a 97.8 % decrease in wear rate. Structural analyses further revealed the presence of screw dislocations, edge dislocations, and slip bands in the USG-treated samples, contributing to the enhanced wear resistance. This study introduces an innovative and effective method to enhance the surface hardness and tribological performance of AISI 201 alloy steel.

1. Introduction

Austenitic stainless steels are primarily classified into two categories: the 300 series, predominantly based on chromium-nickel (Cr-Ni), and the 200 series, based on chromium-manganese-nickel (Cr-Mn-Ni). The 300 series is more expensive due to its higher nickel content. In contrast, the 200 series effectively reduces costs by partially replacing nickel with nitrogen (N) and manganese (Mn) [1–4]. For instance, AISI 201 stainless steel, a typical representative of the 200 series, costs only one-third of 316 stainless steel, while both exhibit comparable corrosion resistance [5–8]. However, AISI 201 stainless steel has relatively low hardness and wear resistance, limiting its extensive application in industrial sectors. Therefore, exploring effective methods to enhance the surface hardness and wear resistance of 200 series stainless steels is of paramount importance [9–11].

There is a wide variety of surface engineering techniques aimed at enhancing the hardness and wear resistance of materials. These techniques modify the material by altering the surface composition and microstructure, primarily including shot peening, physical vapor deposition (PVD), thermal spraying, and heat treatment. Although each of these techniques has its advantages, they also present certain limitations [12–16]. For example, PVD and Chemical Vapor Deposition (CVD) techniques are primarily used to fabricate thin films that are several micrometers thick; however, they are relatively costly [17,18]. In addition, Cheng-Hsun Hsu et al. [19] prepared Ti-N-O coatings via cathodic arc deposition technology, which reduced the wear rate of stainless steel from 13.8×10^{-6} to 1.2×10^{-6} g/m. S. Muthukumar et al. [20] improved the wear resistance of AISI 1045 steel by 30 % using plasma spraying technology. Yu Gao et al. [21] enhanced the wear resistance of 316L steel by eight times using laser cladding technology. However, these methods generally face issues such as high cost, complex equipment requirements, and technical difficulties, which limit their widespread application. In contrast, ultrasonic shot peening (USP), a commonly used surface strengthening technique, offers advantages such

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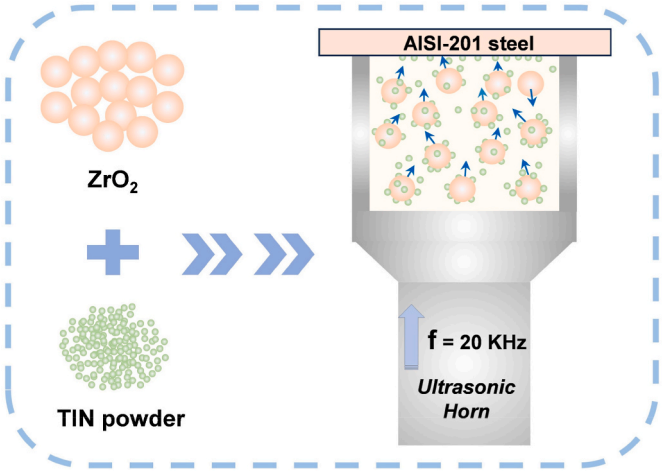


Fig. 1. Schematic of the USG processing method.

as low cost, high efficiency, and ease of operation. The strengthening mechanism involves inducing intense plastic deformation on the surface through strong impacts, thereby enhancing the strength and hardness [22–24]. However, traditional ultrasonic shot peening technology does not alter the material's surface composition, thereby limiting its enhancement effects [25]. To address this limitation, researchers have explored the incorporation of particles into the USP process. Li et al. [26] introduced niobium (Nb) particles into the ultrasonic shot peening process, successfully fabricating Nb coatings on the surface of Inconel 718 alloy. The volume wear rate of the USP-Nb-Aged sample was reduced by 68.9 %. Li et al. [27] demonstrated that the introduction of Al_2O_3 particles into the ultrasonic shot peening process successfully deposited wear-resistant layers on the surface of 18CrNiMo7-6 steel, resulting in a reduction of the friction coefficient and wear rate by 14.4 % and 58.3 %, respectively. However, current research has mainly focused on surface strengthening of high-strength alloys, while there has been relatively little research on materials with lower hardness, such as stainless steels. AISI 201 stainless steel, being a cost-effective and widely used material, has significant potential for improvement in wear resistance. Meanwhile, titanium nitride (TiN) particles possess excellent comprehensive properties, significantly enhancing wear resistance due to their high hardness, while further reducing the friction coefficient and wear rate during friction through chemical reactions [28,29]. However, the application of TiN particles in the USP process and its strengthening mechanisms have not been extensively studied.

To address this research gap, this study introduces TiN particles into the USP process to enhance the surface properties of AISI 201 stainless steel. Herein, TiN powder was incorporated into the ultrasonic shot peening process to fabricate TiN coatings on the surface of AISI 201 alloy steel. Our experiments have demonstrated that the USG method enhances the hardness and tribological properties of AISI 201 alloy steel, while detailed analysis elucidates the mechanism underlying the improved wear resistance. To achieve this, a series of experiments, including microhardness testing, friction and wear testing under dry sliding conditions, X-ray diffractometer phase analysis, Transmission Electron Microscopy microstructure characterization, scanning electron microscope analysis, 3D microscope surface observation, XPS surface chemical composition analysis, among others, were conducted to systematically evaluate the properties of the USG-treated samples. This research results provide new insights for optimizing surface strengthening techniques and demonstrate their considerable potential in industrial applications.

2. Materials and methods

2.1. Sample preparation

In this work, ultrasonic shot peening technology was employed to deposit a reinforced modification layer on the sample surface [26,30,31]. Briefly, TiN powder with an average particle size of approximately 75 μm was used. The TiN powder was mixed with zirconia balls in a 4:1 ratio to create TiN grinding media. The mixed grinding balls were placed in the ultrasonic processing chamber, with the sample fixed at the top of the chamber. The ultrasonic vibration frequency was set to 20 kHz, and the grinding balls were driven to impact the sample surface for surface reinforcement treatment (termed USG sample). The treatment time was set to 3 min. A schematic of the processing setup is shown in Fig. 1 To compare the effectiveness of surface reinforcement, the experiment included a control group. The control group samples were polished but not subjected to any surface reinforcement and are denoted as Base samples. Another set of samples underwent ultrasonic shot peening using only zirconia balls after polishing (named USP sample).

2.2. Characterization

The surface morphology, microstructure, and wear surface were analyzed using a Micro 1000 3D microscope and a TESCAN MIRA LMS scanning electron microscope (SEM, Czech Republic) equipped with an energy dispersive spectrometer (EDS). Cross-sectional microstructure and elemental composition were examined with a focused ion beam scanning electron microscope (FIB-SEM, Helios G4 PFIB) and transmission electron microscopy (TEM). The surface roughness (Ra) of the samples was measured using a surface roughness tester (XM-200, Luoyang Bearing Research Institute Co., Ltd., China). Each sample was tested at three different locations, and the average value and standard deviation were calculated. The Vickers hardness distribution across the cross-section was measured using an HV-1000 tester with a 0.3 N load for 10 s. The measurements were repeated three times at the same cross-sectional depth, and the average value and standard deviation were calculated. The phase composition and chemical states of the sample were analyzed using an X-ray diffraction (XRD) instrument (Rigaku SmartLab SE) under Cu Kα radiation conditions. The elements and their chemical valence states were characterized using a Thermo Scientific K-Alpha X-ray photoelectron spectrometer (XPS). A monochromatic Al Kα source (1486.6 eV) was employed as the excitation source, with a base pressure better than 10^{-7} mBar, an electron emission angle of 90°, and an analysis area of 400 μm^2. The survey spectra were scanned with a step size of 1.0 eV, and high-resolution spectra for Ti 2p, O 1s, and Fe 2p were collected with a step size of 0.1 eV [32,33]. Prior to analysis, the samples were not sputtered, and their surfaces may have contained adsorbed carbon compounds and environmental oxygen contaminants. Therefore, a calibration method based on the sample's work function (Φ_{SA}) was used to ensure the reliability of the results, where the C 1s binding energy is derived from 289.58 − Φ_{SA}. Here, Φ_{SA} represents the work function of the sample [34–38]. Since the work function of TiN is 4.78 eV [39], the calibration was performed using the C 1s standard peak at 284.80 eV.

2.3. Dry sliding test

Tribological testing was conducted using the CFT-I tribometer manufactured by Lanzhou Zhongke Kaihua Technology Development Co., Ltd., employing the rotating ball-on-disk dry friction mode [40–42]. GCr15 bearing steel, chosen for its high hardness and excellent wear resistance [43], was selected as the counterface ball material, with a ball diameter of 6 mm. The tests were conducted at room temperature (25 ± 3 °C), with load ranges of 2 N, 5 N, and 8 N, and sliding speed ranges of 100 r/min, 150 r/min, 200 r/min, 250 r/min, and 300 r/min. To ensure the reliability of the results, each set of sliding tests was repeated at least

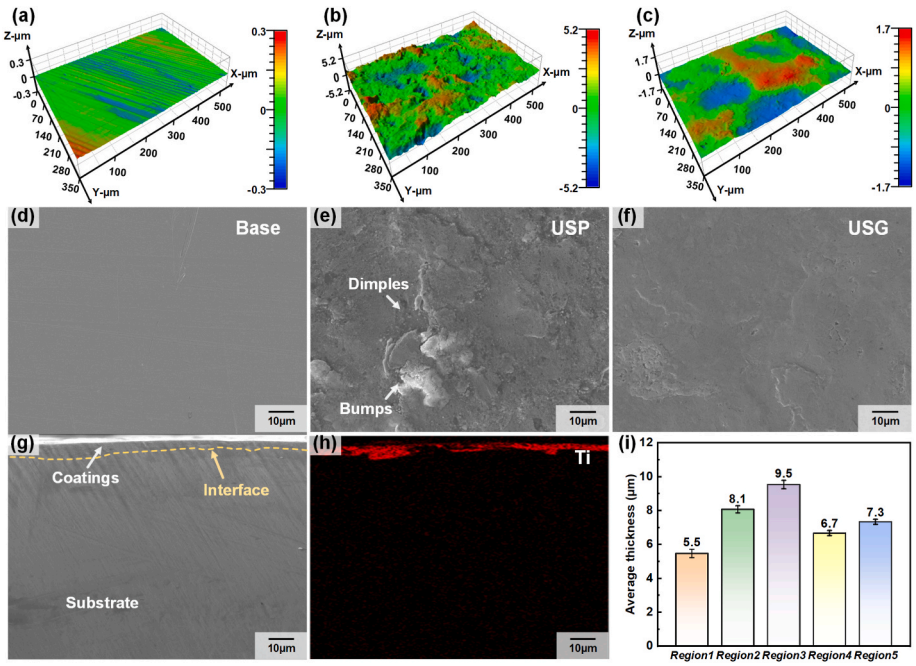


Fig. 2. Surface cross-sections of the Base, USP, and USG samples: (a–c) 3D surface morphology maps of the three samples; (d–f) surface SEM images of the three samples; (g–h) cross-sectional morphology and corresponding EDS spectra of the USG sample; (i) Average thickness distribution of the TiN coating in different regions.

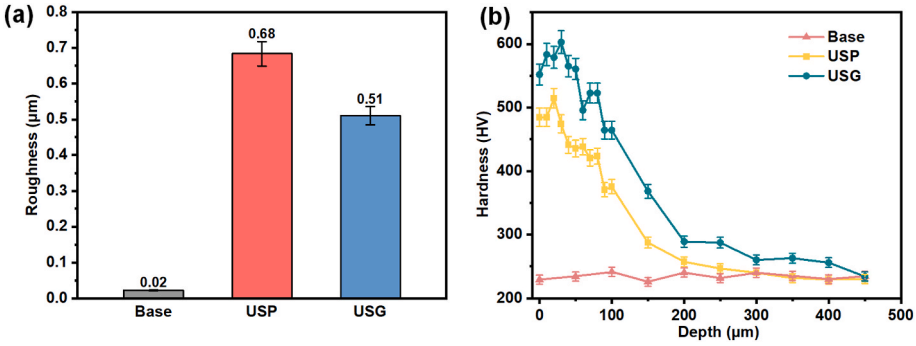


Fig. 3. (a) Comparison of surface roughness for the three samples; (b) Comparison of cross-sectional microhardness for the three samples.

three times, and the average value and standard deviation were calculated. The experimental duration was set to 15 min, and the entire loading and operation process was automatically controlled by sensors and a computer, with real-time recording of pressure and the COF. The wear rate (Wr) of the samples was calculated using the formula $Wr = V/(F \times S)$, where V is the wear volume (mm^3), F is the normal load (N), and S is the sliding distance (m) [44].

3. Results and discussion

3.1. Surface morphology and microstructure

Fig. 2 presents the surface morphology and elemental distribution of the Base, USP, and USG samples. The 3D surface morphology analysis of Fig. 2a indicates that the Base sample has a flat and smooth surface with low roughness. After USP treatment, the surface roughness increases markedly, exhibiting noticeable protrusions and depressions (Fig. 2b). This indicates that the high-frequency impacts of the shot peening balls alter the surface morphology. In contrast, the surface roughness of the USG-treated sample (Fig. 2c) increases but to a lesser extent, remaining substantially lower than that of the USP sample.

The SEM image in Fig. 2d further shows that the Base sample has a smooth surface. Fig. 2e shows the USP-treated sample, which displays distinct pits and protrusions on its surface. In contrast, the surface of the USG-treated sample shown in Fig. 2f is more uniform, with minimal variation. Fig. 2g shows the microstructure of the coating and substrate interface for the USP-treated sample, with a coating thickness of approximately 5–10 μm. The interface is clear, indicating a strong bond between the coating and the substrate without any noticeable separation. Fig. 2h shows the distribution of Ti elements within the coating, revealing the transition of elements at the interface and indicating that the coating is uniformly applied and well-bonded. To evaluate the impact of coating thickness on performance, multiple region thickness measurements were conducted, as shown in Fig. 2i. The coating thickness is uniformly distributed across different regions, with a thickness range from 5.5 μm to 9.5 μm, ensuring effective coverage of the sample surface.

Fig. 3a shows that the base sample exhibits a surface roughness of Ra 0.02 μm. After USP treatment, After USP treatment, the surface roughness increases to Ra 0.68 μm due to the high-energy shot peening balls creating indentations and protrusions. In contrast, USG treatment, with the introduction of TiN particles, reduces the surface roughness to Ra

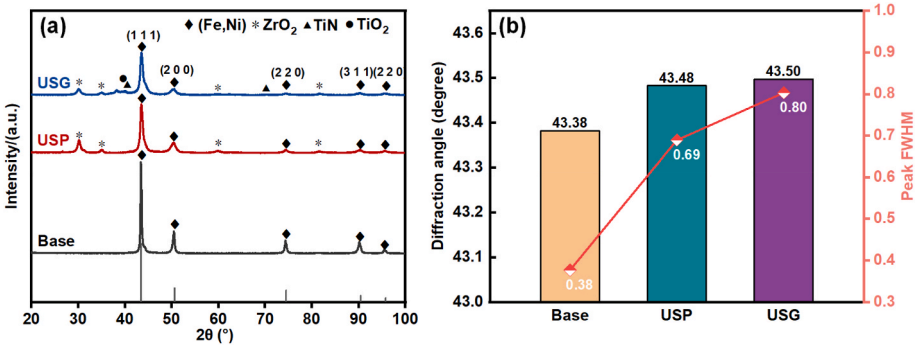


Fig. 4. (a) XRD diffraction patterns of the three sample groups; (b) Diffraction angles and full width at half maximum (FWHM) for the (111) crystal plane index.

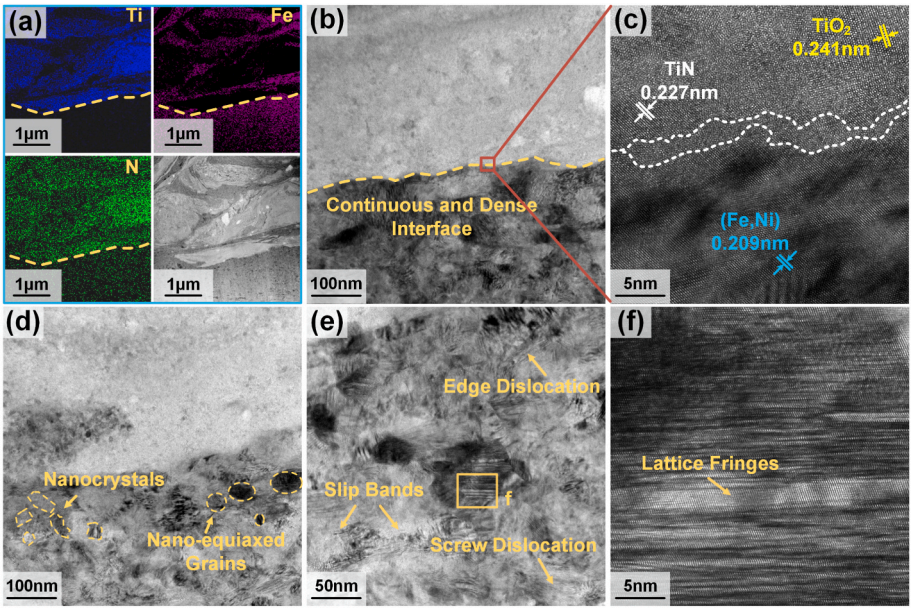


Fig. 5. Coating-substrate interface cross-section: (a) SEM image of the interface and EDS elemental analysis; (b, c) HRTEM image; (d–f) High-resolution (HRTEM) images of the substrate.

0.51 μm, resulting in a smoother surface compared to the USP-treated sample. Fig. 3b presents the trend of hardness variation with depth in the sample cross-section. The average hardness of the Base sample is 234.3 HV. Following USP treatment, the surface hardness increases to approximately 514.7 HV, representing an improvement of 120 %. However, with increasing depth, the hardness decreases rapidly, approaching that of the Base sample around 300 μm. In contrast, the surface hardness of the USG-treated sample reaches a maximum of 603.0 HV, reflecting a 157 % increase compared to the Base sample. At a depth of 300 μm, the hardness of the USG-treated sample remains superior to that of the USP-treated sample. The hardened layer depth of the USG-treated sample reaches approximately 450 μm, surpassing that of the USP-treated sample. This suggests that USG treatment not only enhances surface hardness but also maintains higher hardness at greater depths, demonstrating a more uniform and enduring hardening effect.

Fig. 4a compares the XRD diffraction patterns of the three sample groups. The main diffraction peaks of the Base sample correspond to the (111), (200), and (220) crystal planes of the (Fe, Ni) crystal structure, indicating its typical face-centered cubic (FCC) structure [45]. In contrast, in the USP sample, in addition to the original (Fe, Ni) diffraction peaks, a new diffraction peak corresponding to ZrO_2 is clearly observed. For the USG samples, characteristic diffraction peaks of TiN and TiO_2 were detected. The presence of TiN indicates that during the USG process, TiN particles were successfully embedded into the matrix surface through ultrasonic shot peening, accompanied by a certain degree of oxidation to form TiO_2. The multiphase structure formed by the interaction of these TiO_2 and TiN phases significantly enhances the hardness and wear resistance of the material [46]. Fig. 4b shows the trend of changes in the diffraction angle and peak width (FWHM) of the (111) plane for all three samples. The diffraction angle of the Base sample is 43.38°, with an FWHM of 0.38. After USP treatment, the diffraction angle shifts to 43.48°, and the FWHM increases to 0.69. The USG treatment further shifts the diffraction angle to 43.50°, with the FWHM increasing to 0.80. This indicates that both USP and USG treatments cause lattice distortion in the samples, leading to an increase in dislocation density and residual stress, thereby resulting in a broader peak and a shift of the diffraction angle to the right [47,48]. Based on XRD spectra and the Williamson-Hall method [49] (see Supplementary materials), the calculated dislocation density for the Base, USP, and USG samples is 3.16×10^{14} m^{-2}, 19.99×10^{14} m^{-2}, and 21.90×10^{14} m^{-2}, respectively. It can be observed that the dislocation density of the USP and USG samples after treatment is significantly higher than that of the Base sample. A higher dislocation density is typically accompanied by stronger crystal defects and microstrain, which enhances the material's resistance to deformation and its hardness [50].

Fig. 5 presents the microstructural and elemental distribution analysis of the samples conducted via transmission electron microscopy. The low-magnification TEM images in Fig. 5a clearly depict the interface

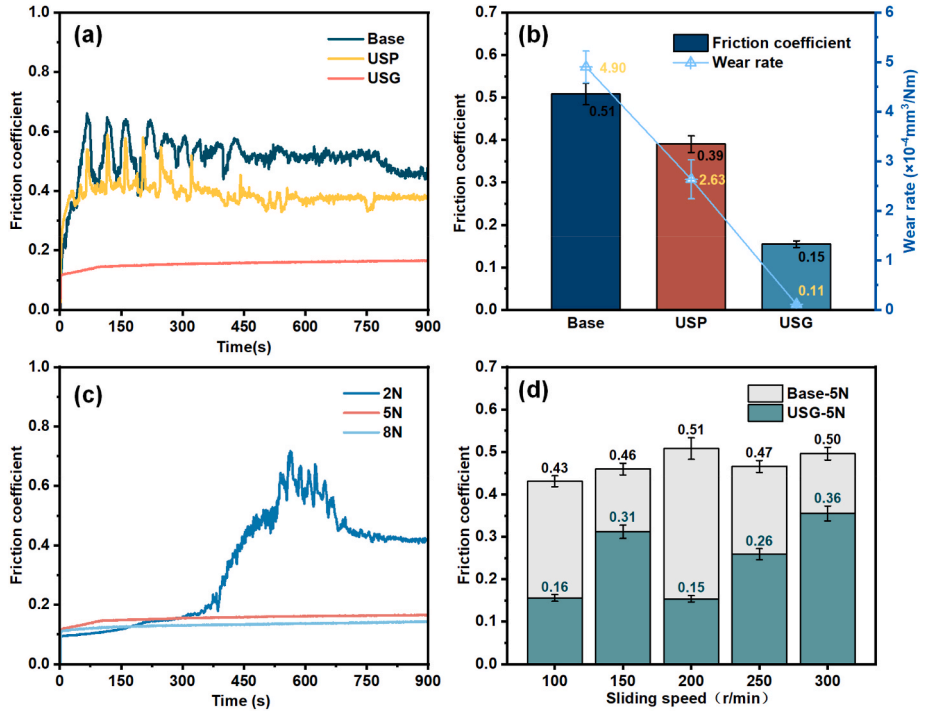


Fig. 6. (a–b) Comparison of COF and wear rate for three samples under experimental conditions of 5 N and 200 r/min; (c) COF curves of the USG sample under different loads; (d) Comparison of average COF between the Base and USG samples at different rotational speeds.

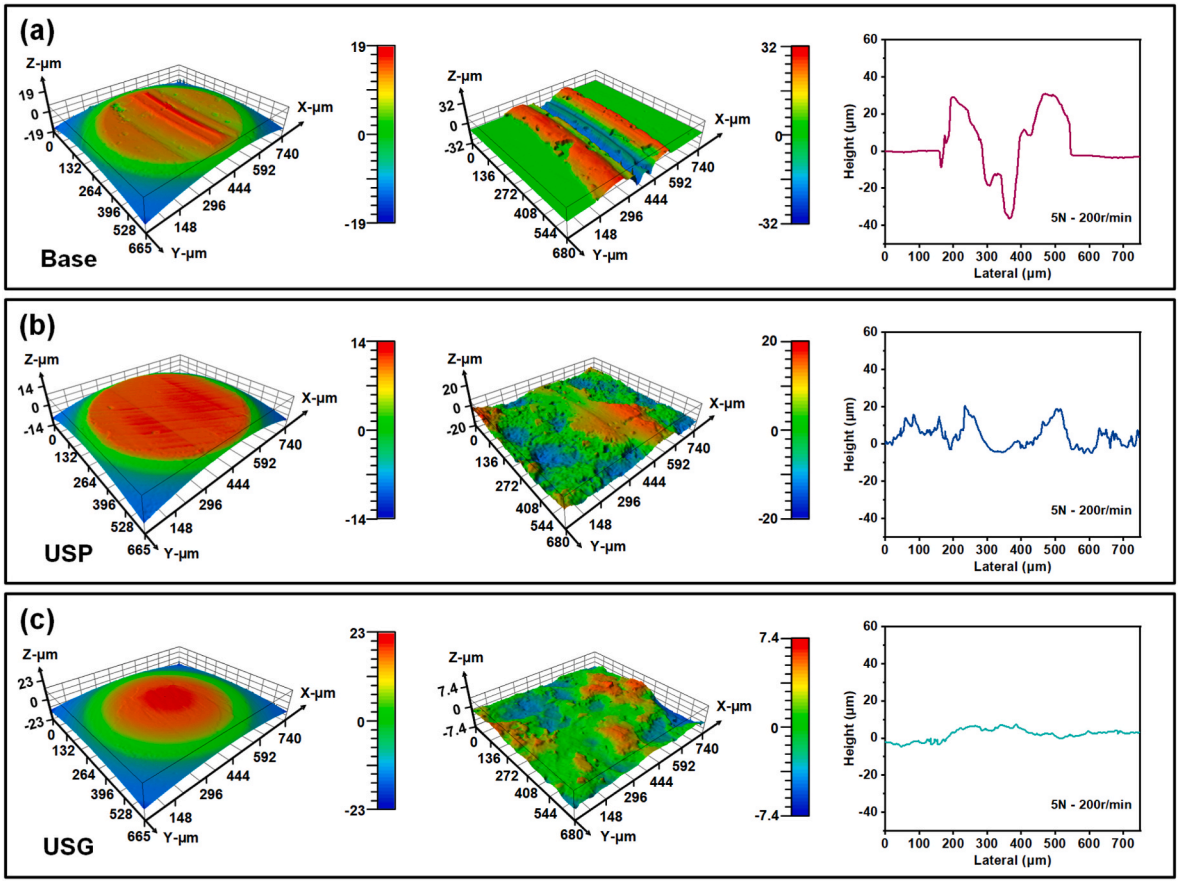


Fig. 7. 3D morphology of wear tracks and depth profiles for the three samples and their counterface balls under 5 N and 200 rpm conditions: (a) Base sample; (b) USP sample; (c) USG sample.

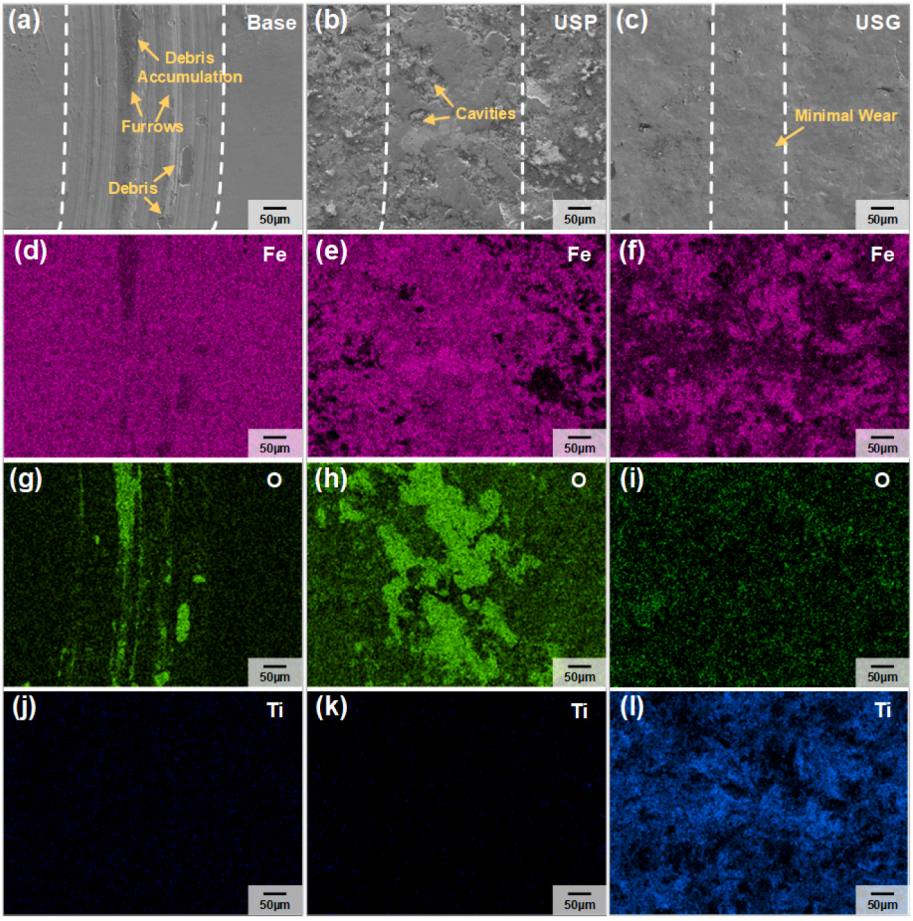


Fig. 8. SEM images of wear tracks and their corresponding EDS spectra for the three samples tested under 5 N and 200 rpm conditions. (a) Base sample SEM image with EDS spectra (d), (g), and (j); (b) USP sample SEM image with EDS spectra (e), (h), and (k); (c) USG sample SEM image with EDS spectra (f), (i), and (l).

between the TiN coating and the substrate. The results indicate that the TiN coating is primarily composed of Ti and N, with uniform and well-defined elemental distribution. Fig. 5b and c presents high-resolution TEM (HRTEM) images of the TiN coating and substrate interface. The lattice spacing of the TiN coating is 0.227 nm, the lattice spacing of the TiO_2 layer is 0.241 nm, and the lattice spacing of the matrix (Fe, Ni) phase is 0.209 nm. The images reveal a clear interface structure between the coating and the substrate, indicating a strong bond and stability at the interface. Furthermore, the TiN particles at the coating interface act as grain boundary pinning agents, restricting the relative motion of the matrix grains, thus enhancing the strength and stability of the samples [51].

Fig. 5d demonstrate pronounced grain refinement in the USG-treated sample, forming uniformly distributed nanograins, including equiaxed nanocrystals and irregularly shaped nanoparticles [52]. USG treatment refines the surface grains, resulting in a nanocrystalline structure. For polycrystalline materials, the relationship between strength/hardness and grain size typically follows the Hall-Petch equation [53]:

$$\sigma_y = \sigma_0 + Kd^{-1/2} \quad (1)$$

Where σ_y is the yield stress of the material,σ_0 is the lattice friction resistance during single dislocation motion, K is a constant, and d is the average grain diameter. Typically, the yield stress or hardness of the material is proportional to $d^{-1}/^2$ indicating that reducing the grain size significantly enhances hardness. Grain refinement, through the Hall-Petch effect, improves hardness, while the dislocation proliferation effect further strengthens the surface layer's resistance to deformation [54].

Fig. 5e shows the presence of screw dislocations, edge dislocations, and slip bands in the coating, indicating that USG treatment promotes the proliferation of dislocations within the material [55]. These crystal defects strengthen the material's resistance to deformation by restricting further dislocation motion, which alters the surface contact conditions and improves the material's frictional performance [56]. Despite the proliferation of dislocations, Fig. 5f still show uniform and distinct lattice fringes, indicating that USG treatment preserves the internal crystal order.

3.2. Tribological performance

Fig. 6 show the variation of COF with time for the three groups of samples. In Fig. 6a, the curves of COF variation with time show that the Base sample exhibits a higher and more fluctuating COF, which can be attributed to its lower wear resistance and less stable surface characteristics [57,58]. After USP treatment, the COF decreases due to surface hardening [59]. Notably, the USG sample exhibits the lowest and most stable COF, indicating that the synergistic effect of TiN particles and ultrasonic shot peening plays a key role in improving wear resistance [60]. Fig. 6b quantifies the average COF of the three samples, with values of 0.51, 0.39, and 0.15 for the Base, USP, and USG samples, respectively. Compared to the Base sample, the COF of the USG sample decreased by 70.6 %, and compared to the USP sample, it decreased by 60.5 %. Moreover, the wear rate of the Base sample was 4.90×10^{-4} mm^3/Nm, while the wear rate of the USP sample decreased to 2.63×10^{-4} mm^3/Nm, a reduction of 46.3 % compared to the Base sample. The wear rate of the USG sample further decreased to 0.11×10^{-4} mm^3/Nm, a reduction of 97.8 % compared to the Base sample. These results

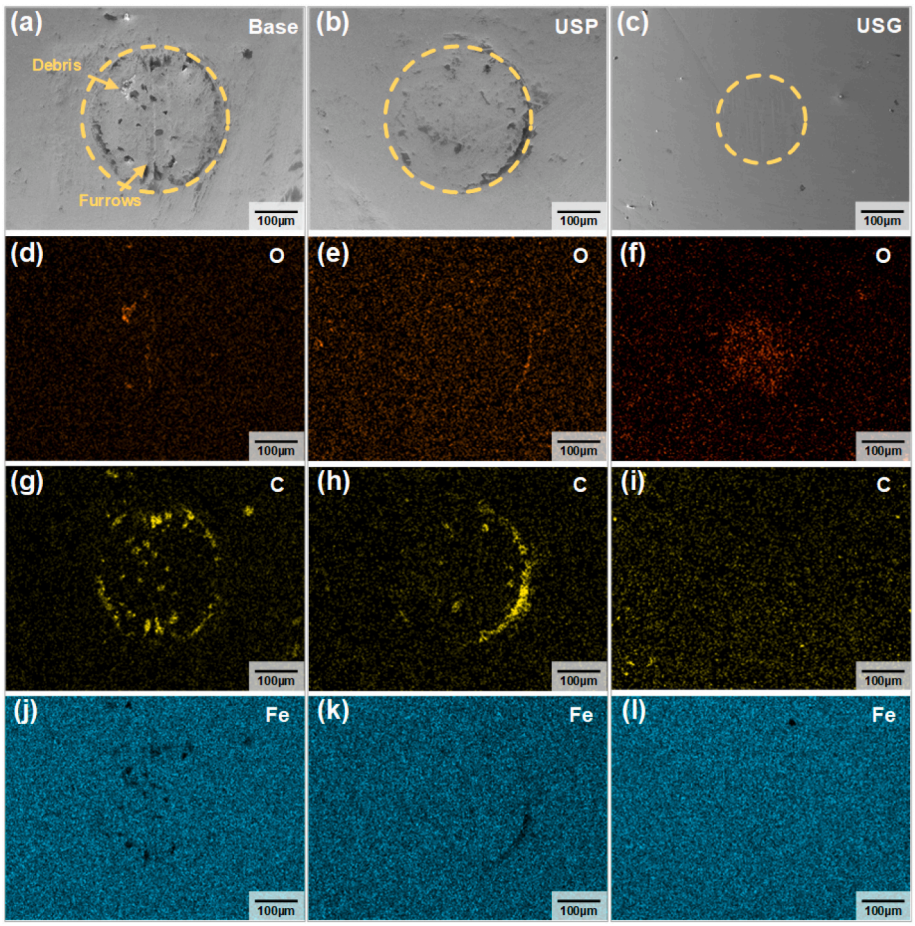


Fig. 9. SEM images and corresponding EDS spectra of wear tracks on counterface balls for three samples: (a) Base sample SEM image with EDS spectra (d, g, j); (b) USP sample SEM image with EDS spectra (e, h, k); (c) USG sample SEM image with EDS spectra (f, i, l).

demonstrate the significant effectiveness of USG treatment in reducing both the COF and wear rate.

Fig. 6c shows the variation of the COF for the USG sample under different loads. Under applied loads of 2 N and 5 N, the COF remains low and stable. However, under an 8 N load, the COF remains relatively steady for the first sliding distance of approximately 18.85 m, then rapidly increases between 18.85 and 43.98 m. Fig. 6d compares the COF of the Base and USG samples at various rotational speeds. Regardless of the rotational speed, the COF of the USG-treated sample exhibits a lower COF than the Base sample, with minimal variation. This demonstrates that the TiN coating exhibits excellent anti-friction performance across various rotational speeds. In contrast, the COF of the Base sample increases with rotational speed, indicating poorer tribological performance. In summary, the USG process enhances the frictional properties of AISI 201 steel by forming a TiN coating on its surface.

3.3. Wear track morphology analysis

Three-dimensional wear track and cross-sectional analyses clearly reveal the effects of different surface treatment processes on the wear resistance of AISI 201 alloy steel. Fig. 7a shows the wear track morphology of the Base sample, which exhibits deep grooves, severe surface wear, and noticeable material accumulation. The cross-sectional wear curve shows a maximum depth of 36.3 μm, indicating that the Base sample has low surface hardness, poor tribological performance, and experiences significant material removal and plastic deformation during friction. In contrast, Fig. 7b shows the wear track of the USP-treated sample after shot peening, where the wear track depth is markedly reduced. The three-dimensional morphology reveals a smoother surface, with a maximum depth of only 11.9 μm. This suggests that the shot peening process effectively enhances anti-wear properties and reduces material loss. Fig. 7c presents the wear track of the USG sample, which demonstrates the best sliding condition. Its three-dimensional morphology shows the shallowest and most uniform wear marks, with virtually no noticeable signs of wear. The TiN coating enhances surface hardness, effectively reducing plastic deformation and material removal. Under high-load conditions, the USG treated sample maintains excellent stability and wear resistance, demonstrating the best anti-wear performance.

Fig. 8 presents the SEM images of surface wear tracks and the corresponding EDS elemental spectra of three samples, offering insights into the micro-wear behavior influenced by different surface treatment methods. Fig. 8a presents the SEM image of the Base sample, clearly showing grooves, material accumulation, and debris, indicating significant material removal during the friction process, and suggesting that the sample has undergone severe abrasive wear. The EDS elemental spectrum (Fig. 8d) reveals a uniform distribution of Fe in the wear zone, suggesting extensive substrate exposure, while the accumulation of O in the wear track (Fig. 8g) indicates localized oxidative wear due to high temperatures generated during friction.

Fig. 8b shows the SEM image of the USP-treated sample, revealing a wear track that is noticeably smoother than that of the Base sample. However, small pits caused by the shot peening process are visible, which may result in fluctuations in the COF during friction. The absence of significant plowing grooves suggests that the USP treatment effectively mitigates wear by enhancing surface hardness. The EDS elemental mapping (Fig. 8e) shows a reduction in Fe exposure, indicating that less substrate material is exposed. Furthermore, the localized enrichment of

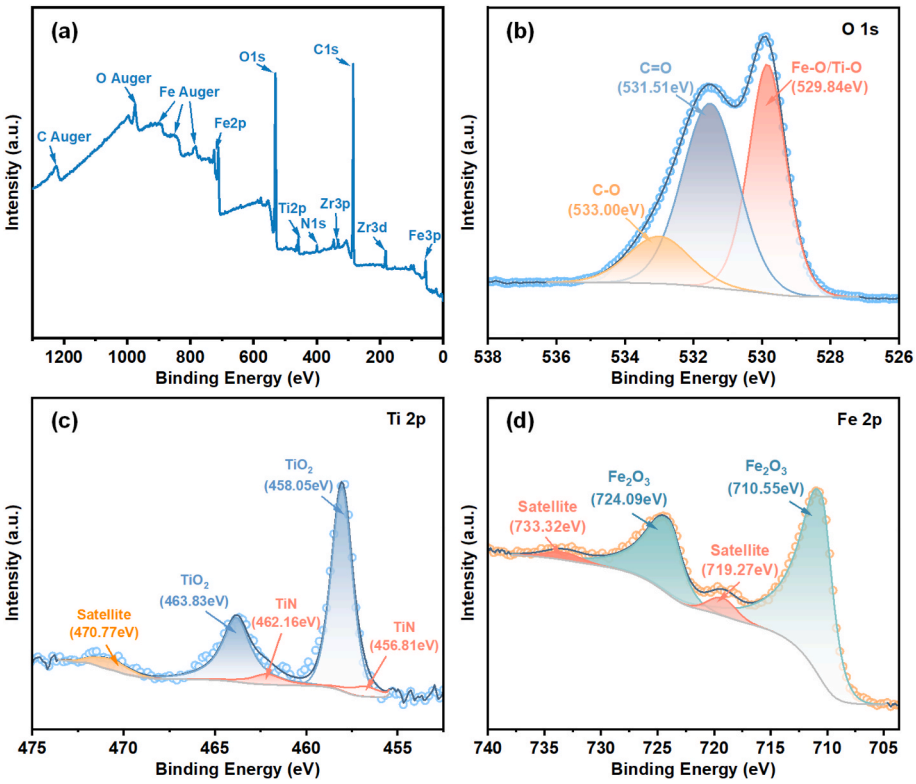


Fig. 10. XPS spectra of the wear track on the USG sample: (a) XPS full-spectrum; (b), (c), and (d) high-resolution XPS spectra of O1s, Ti2p, and Fe2p for the USG sample, respectively.

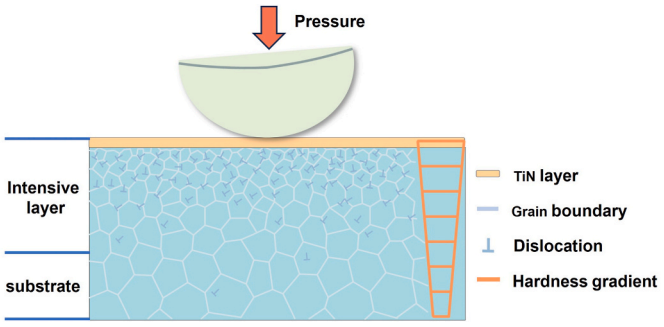


Fig. 11. Schematic of the tribological mechanism of the USG sample.

oxygen (Fig. 8h) suggests the formation of an oxide layer, primarily due to oxidative wear. This oxide layer may act as a lubricating film, reducing direct contact between the sample and the counterface ball, thereby contributing to the reduction of wear and friction during the sliding process [61].

Fig. 8c illustrates the SEM image of the USG-treated sample, where minimal abrasive wear is observed, with no visible plowing grooves or material accumulation. The smooth surface morphology indicates a substantial improvement in wear resistance compared to the Base and USP samples. The EDS elemental mapping (Fig. 8f) reveals a further reduction in Fe exposure, while the uniform distribution of Ti (Fig. 8j) confirms that the TiN coating acts as a protective layer, effectively preventing substrate exposure and further wear. Furthermore, under high-temperature conditions, the wear behavior of stainless steel coatings may be significantly influenced by the formation of an oxide film. For example, the decrease in the friction coefficient of TiN/TiO_2 composite coatings at high temperatures is primarily attributed to the formation of the oxide film, rather than solely to the increase in coating hardness [62]. These results demonstrate that the USG-treated sample exhibits markedly reduced wear, underscoring the crucial role of the TiN coating in enhancing the tribological performance of AISI 201 alloy steel.

Fig. 9 presents the SEM images of wear tracks on the counterballs and the corresponding EDS elemental spectra for the three surface treatment methods, providing insights into the effects of different treatments on the counterballs' wear behavior during friction. Fig. 9a shows the SEM image of the Base sample. Plowing grooves and wear debris are observed in the wear track, indicating material removal during the friction process. The EDS spectrum (Fig. 9d) reveals local accumulation of O in the wear track, emphasizing significant oxidation. Meanwhile, C elements are concentrated in the wear region (Fig. 9g), and a noticeable migration of Fe further indicates the exposure and transfer of substrate material due to wear. Fig. 9b illustrates the SEM image of the USP-treated sample, which shows pronounced improvement in wear behavior. The wear track is shallower with notable reduced wear marks. The EDS spectrum (Fig. 9e) reveals a broader distribution of O, suggesting oxidation as the dominant wear mechanism. C elements are concentrated at the outer edges of the wear track (Fig. 9h), indicating more localized carbon transfer. The migration of Fe is still observed (Fig. 9k), though its extent has decreased compared to the Base sample. Fig. 9c shows the SEM image of the USG-treated sample, demonstrating the highest wear resistance. The wear track is nearly imperceptible, with only minimal wear marks. The EDS spectrum (Fig. 9f) reveals a reduced presence of O, indicating minimal oxidation wear. The C elements are uniformly distributed at a low concentration (Fig. 9i), while the distribution of Fe (Fig. 9l) suggests negligible substrate material migration. These findings demonstrate that the USG-treated sample, with surface reinforcement, is highly effective in minimizing material removal and reducing counterball wear. The TiN coating provides exceptional protection, substantially reducing wear and highlighting the significant advantages of USG treatment in enhancing wear resistance.

Fig. 10 presents the XPS analysis of the elemental composition and chemical states on USG-treated sample following friction. The survey

spectrum (Fig. 10a) identifies elements such as C, O, Fe, Mn, Ti, Zr, and N, confirming their presence on the sample surface following friction. High-resolution analysis in Fig. 10b shows the O 1s spectrum, confirming Ti-O and Fe-O (529.84 eV) on the surface. The Ti 2p spectrum in Fig. 10c further confirms the presence of TiO_2 (458.05 eV, 463.83 eV) and TiN (456.81 eV, 462.16 eV) [63,64], indicating that these protective layers remain stable during friction, thereby reducing wear. The Fe 2p spectrum in Fig. 10d reveals Fe_2O_3 (710.55 eV, 724.09 eV) along with its satellite peaks [65], suggesting Fe oxidation and the formation of an iron oxide layer during friction. The XPS results demonstrate that the TiO_2 and TiN layers on the USG-treated sample remained stable during friction, effectively protecting the surface and minimizing wear.

3.4. Anti-wear mechanism

Fig. 11 illustrates a multi-layered structure formed on the surface of AISI 201 alloy steel. The reinforcement layer comprises the TiN coating and a gradient layer characterized by grain refinement and dislocations. The TiN coating, as a surface-strengthening component, significantly enhances surface hardness and tribological performance [66]. During friction, the coating absorbs and effectively disperses external pressures, reducing direct wear. The strong adhesion between the coating and substrate ensures the stability and wear resistance of the material under high-stress and frictional conditions. Simultaneously, the gradient structure beneath the coating, formed via ultrasonic shot peening, exhibits an equiaxed nanocrystalline structure, accompanied by abundant screw dislocations, edge dislocations, and slip bands. These crystal defects hinder dislocation movement and suppress plastic deformation, further enhancing material hardness and anti-wear properties [67]. The hardness gradient within the reinforcement layer gradually distributes and transfers external stresses, reducing stress concentration and minimizing the risk of surface failure [68,69]. Ultimately, the combined effect of the coating and the gradient-enhanced structure significantly improves to the wear resistance of AISI 201 alloy steel, thereby extending its service life.

4. Conclusion

This study explores the underlying wear-resistance mechanisms by incorporating TiN particles into the ultrasonic shot peening process, revealing significant enhancements in its tribological performance.

1) The USG treatment incorporates TiN particles into the ultrasonic shot peening process, leading to a substantial improvement in the microhardness of the sample, which increases by 157 % compared to the untreated Base sample.
2) Tribological testing shows that the USG-treated sample exhibits a significantly lower and more stable COF, with the wear volume reduced by 97.8 % compared to the Base sample. This is primarily attributed to the combined effects of surface hardening, grain refinement, increased dislocation density, and the synergistic role of TiN particles in reducing friction.
3) These findings provide a novel approach to enhance the surface hardness and wear resistance of AISI 200-series alloy steels, contributing to the development of materials with superior tribological properties.

CRediT authorship contribution statement

Haobing Hu: Writing – original draft, Visualization, Methodology, Formal analysis. **Ke Chen:** Software, Investigation, Data curation. **Fenghua Liu:** Methodology, Formal analysis. **Yiting Pan:** Methodology, Formal analysis. **Meicong Wang:** Investigation, Formal analysis. **Zhongwei Liang:** Supervision, Methodology. **Gongbin Tang:** Writing – review & editing, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.vacuum.2025.114180.

Data availability

Data will be made available on request.

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